

KENTECH Career Lab

Product Requirements Document

Introduction to AI Programming · KENTECH · Spring 2026

Assignment 4 Submission

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GitHub: <https://github.com/ogidaa0522/kentech-career-lab>

Deployment: <https://kentech-career-lab.vercel.app>

1. Product Overview

KENTECH Career Lab is an AI-powered career guidance platform built for students at Korea Institute of Energy Technology (KENTECH). It helps students prepare for job interviews in the energy sector and explore personalized career paths — including employment at public corporations, private companies, research institutes, and graduate school admission.

Unlike generic career platforms, KENTECH Career Lab is specialized for energy engineering majors, covering fields such as renewable energy, nuclear power, hydrogen, smart grid, battery/ESS, and energy policy. All AI features are powered by the Claude API (Anthropic), running entirely in the browser without a backend server.

2. Target Users

Primary Users

- Undergraduate students at KENTECH (all years)
- Students preparing for job applications at energy companies or public corporations
- Students considering graduate school (domestic or overseas)
- Students unsure about their career direction in the energy field

User Needs

- Tailored interview question practice for energy sector jobs
- Concrete and personalized career path recommendations
- Guidance specific to KENTECH's energy engineering curriculum
- An easy-to-use tool that requires no account or setup

Pain Points

- Generic career platforms do not cover energy-specific roles
- Interview questions online are not tailored to Korean energy companies

- Students lack structured guidance on graduate school vs. employment
- No platform addresses KENTECH's unique interdisciplinary energy curriculum

3. Project Goals

- **Goal 1:** Provide AI-generated, energy-sector-specific interview questions for Korean companies and research institutes.
- **Goal 2:** Offer personalized career path recommendations covering employment, graduate school, and overseas options.
- **Goal 3:** Build a clean, responsive, and instantly usable web service with no login or backend required.
- **Goal 4:** Demonstrate integration of the Claude API (Anthropic) as a core product feature.
- **Goal 5:** Complete the full product development cycle: PRD → implementation → Vercel deployment.

4. Core User Scenario

Persona: Jiwon, a third-year KENTECH student preparing to apply for an internship at KEPCO (■■■■■■■).

- **Step 1:** Jiwon visits KENTECH Career Lab on her laptop.
- **Step 2:** She goes to the Interview page and selects '■■■■■ ■■■■■' as the job field and '■■■■■■■ (KEPCO)' as the company.
- **Step 3:** She selects '■■■+■■■ ■■■' interview type and enters additional context about her senior design project.
- **Step 4:** She clicks '■ AI ■■ ■■ ■■■■■' and receives 8 tailored interview questions within seconds.
- **Step 5:** She reads through the questions and copies them to practice later.
- **Step 6:** She then visits the Career page and inputs her interests and strengths.
- **Step 7:** AI recommends 3 career paths — KEPCO employment, KIER research institute, and a domestic master's program — each with a concrete first action step.

Outcome: Jiwon has a clear interview prep list and a confirmed career direction, all within 10 minutes.

5. Feature List

Feature	Description	Priority
AI Interview Question Generator	Generates energy-sector-specific technical and personality interview questions via Claude API, based on job field, company, and interview type	Must-have

AI Career Recommendation	Returns top-3 personalized career paths (employment / grad school / overseas) with reasons, tags, and a first action step	Must-have
Energy Field Specialization	Dropdown menus covering 8 energy job fields and 13 Korean companies, research institutes, and graduate schools	Must-have
Copy to Clipboard	One-click copy of all generated interview questions for offline practice	Should-have
Regenerate Results	Re-call the AI with the same inputs to get a fresh set of questions or recommendations	Should-have
Dark Mode	Toggle between light and dark themes; preference saved in localStorage	Nice-to-have
Responsive Design	Fully usable on mobile and desktop via CSS media queries	Should-have

6. Page Structure

Home (index.html)

- Hero section with service intro and CTA buttons
- Energy career tags strip (KEPCO, KAIST, etc.)
- 3-column feature overview cards
- 3-step 'How It Works' section
- CTA section linking to Interview page

Interview (interview.html)

- Form: job field, company, interview type, question count, extra info
- AI generates questions via Claude API
- Questions displayed as cards with type label and answer hint
- Regenerate and Copy-all buttons

Career (career.html)

- Form: interest field, grade, work style, strength, additional info
- AI returns 3 career recommendations via Claude API
- Each card shows title, description, personal reason, next step, and tags
- Retry button for new recommendations

About (about.html)

- Project and service description
- Core features list

- Tech stack chips
- AI development report (tools, prompts, modifications, bugs)

7. Technical Requirements

Frontend

- HTML5 — semantic page structure
- CSS3 — custom design system, dark mode, CSS variables, responsive layout
- Vanilla JavaScript — API calls, DOM rendering, localStorage

AI / API

- Claude API (claude-sonnet-4-6) — interview question generation and career recommendation

Storage

- Browser localStorage — dark mode preference only; no user data stored

Tools & Deployment

- Visual Studio Code
- GitHub — version control
- Vercel — static site deployment

8. Design Requirements

Visual Style

- Deep navy base (#0F172A) with KENTECH blue (#2563EB) and energy orange (#F97316) accent
- Clean, modern, and professional — appropriate for a career guidance tool
- Orange accent chosen to symbolize energy and dynamism

Typography

- Space Grotesk — headings and logo
- Inter — body text and UI elements

Signature Element

- Animated concentric pulse rings on the hero section, representing energy waves — the single memorable visual element

UX Principles

- No login, no setup — usable immediately
- Clear loading states while AI generates responses

- Error handling with user-friendly messages
- Mobile-responsive layout

9. Milestones

Week	Milestone
Week 1	Service idea selection and PRD completion
Week 1	Wireframe and design system planning
Week 2	HTML/CSS implementation — all four pages
Week 2	Claude API integration for interview question generator
Week 2	Claude API integration for career recommender
Week 2	Testing, bug fixing, and UI polish
Week 2	GitHub repository and README preparation
Week 2	Vercel deployment and final submission